

Data Quality Report — BOAMP Renewal Modeling Dataset

Gigalis Internship — Phase 1 Deliverable L1

Internship Project — Gigalis

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1 Objective and Scope

This report documents the quality of the BOAMP corpus used to construct the renewal proxy and the survival modeling dataset. The scope is the BOAMP-only pipeline for digital-sector public procurement notices in Pays de la Loire over 2015–2024. The official modeling population is the set of eligible APPEL_OFFRE contracts retained in `data/processed/boamp_phase2_survival.csv`. In this report, `event = 1` denotes a *proxy renewal event* found by the linking algorithm; `event = 0` denotes right-censoring or no observed link under the rule before the study end date. It does not prove that no renewal occurred.

2 Raw Corpus Overview

The cleaned BOAMP corpus is stored in `data/processed/boamp_full_clean.csv`. It contains 3,181 notices published between 2015-03-02 and 2024-12-29.

Table 1: Raw BOAMP notice types

Notice type	Count	Share
APPEL_OFFRE	1,933	60.8%
ATTRIBUTION	1,081	34.0%
RECTIFICATIF	150	4.7%
PRE-INFORMATION	7	0.2%
MODIFICATION	5	0.2%
INTENTION_CONCLURE	4	0.1%
PERIODIQUE	1	0.0%
Total	3,181	100.0%

The renewal-linking pipeline uses only the 1,933 APPEL_OFFRE notices as possible source contracts. ATTRIBUTION notices are used as an auxiliary source to refine contract start dates when a BOAMP back-reference is available.

3 Final Modeling Population

The study population is narrower than the raw corpus because not every notice is a candidate source contract and not every source contract has an observable renewal window before the 2024-12-31 study end.

Table 2: Population construction summary

Population	Count	Interpretation
Raw BOAMP notices	3,181	Full digital-sector corpus downloaded from the BOAMP API.
APPEL_OFFRE source notices	1,933	Potential source contracts for renewal linking.
Eligible source contracts	1,100	APPEL_OFFRE contracts whose estimated end date falls before study end.
Linked proxy-renewal events	705	event = 1; a plausible later BOAMP notice was linked.
Right-censored contracts	395	event = 0; no observed link under the rule before study end.

The linking rate in the eligible population is 64.1% (705/1,100). This is the denominator that matters for Phase 2 survival modeling. The lexical baseline retained in the repository for comparison is 11.3%.

4 Missing-Value and Cleaning Summary

Table 3 summarizes the main variable-level quality issues that carry forward into modeling.

Table 3: Variable quality and cleaning decisions

Variable	Issue	Cleaning / treatment	Modeling impact
buyer_siret / buyer_key	Raw buyer_siret coverage is 9.1% in the 3,181-row corpus; post SIREN enrichment, 660 of the 1,100 eligible contracts carry a SIREN: key and 440 a NAME: key.	Build buyer_key as SIREN:... when matched with HIGH confidence via the API Recherche d'Entreprises, else NAME:...	Buyer-name fragmentation is the main source of false negatives in linking; SIREN enrichment partially mitigates this.
CPV fields	cpv_principal is missing for 1.9% of the raw corpus; generic codes reduce specificity. Eligible cpv_div2 coverage is 95.3%.	Clean and standardize CPV; generic codes are capped at division-level similarity; missing CPV is excluded from the composite score with weight renormalization.	Missing or generic CPV weakens auditability and may reduce match discrimination.
amount_clean	Valid amount coverage is 41.1% in the full corpus and 21.4% among APPEL_OFFRE ; 20.3% in the eligible handoff.	Flag zero, tiny, and ceiling values; set flagged amounts to missing.	Amount is too sparse to act as a linking signal and must be handled as optional context only.
declared_duration_months duration_clean	Raw cleaned duration coverage is 47.3% overall and 76.6% for APPEL_OFFRE . In the eligible cohort, 25.1% are imputed.	Remove suspect durations and impute missing eligible durations to 48 months with dur_was_imputed .	Duration affects eligibility and temporal search windows, so imputed rows are less reliable.
start_date	No missing values in the eligible handoff, but 59.3% use publication date and 40.7% use an attribution-derived award date.	Use award date via annonce_lie when available, else publication date.	Temporal scores contain date noise because publication is not always the operational contract start.
type_procedure	0.3% missing in the eligible handoff.	Retain as-is; no imputation required.	Low risk for descriptive summaries and covariate use.
type_marche	14.4% missing in the eligible handoff.	Retain missing values explicitly.	Missingness may blur segment-level interpretation.
event / renewal link	No explicit BOAMP renewal identifier exists; event is algorithmic rather than observed ground truth.	Use the linking algorithm to build a proxy event; keep confidence diagnostics.	False positives and false negatives remain possible, so event quality must be interpreted as bounded evidence.

5 Key Data Quality Issues

The dominant data quality problems are structural rather than cosmetic. Buyer identity was initially weak because the eligible cohort had no SIRET-anchored `buyer_key`; most grouping relied on normalized buyer names. SIREN enrichment (completed 2026-06-24 via the API Recherche d’Entreprises) has partially resolved this: 660 of the 1,100 eligible rows now carry a HIGH-confidence SIREN-enriched buyer identity in `boamp_phase2_survival_siren_enriched.csv`; 440 remain on the name-based key (see Section 11). Residual name-based fragmentation remains possible for the unresolved cases. CPV coverage is high but not perfect, and generic CPV values reduce matching precision. Contract amount is sparse on source notices, so it cannot support renewal linking. BOAMP also has no explicit renewal label, which means the linking output is necessarily an inferred event proxy.

6 Duration Field Reliability

Declared duration is not the survival target. In the `APPEL_OFFRE` population, the cleaned declared duration field is heavily concentrated on administrative values of 12, 24, 36, and 48 months.

Table 4: Declared duration concentration in `APPEL_OFFRE` notices

Duration value	Count	Share of non-missing durations
48 months	462	31.2%
12 months	316	21.4%
24 months	151	10.2%
36 months	99	6.7%
Subtotal exactly 12/24/36/48	1,028	69.5%

This concentration is more consistent with administrative ceilings than with realized renewal timing. For that reason, `declared_duration_months` is used as a covariate and as a temporal search prior, while `observed_duration_months` is the survival time used in Phase 2.

7 Renewal Linking Quality

BOAMP provides no explicit renewal ID, so the renewal variable is constructed by matching later notices from the same `buyer_key` using text, CPV, and timing. The eligible modeling dataset contains 705 linked proxy-renewal events and 395 right-censored contracts.

Table 5: Linking outcome summary

Metric	Value	Interpretation
Linked events	705	Proxy renewal events retained in the official BOAMP hand-off.
Right-censored	395	No observed link before study end under the chosen rule.
Linking rate	64.1%	Share of eligible contracts with <code>event</code> = 1.
Strict high-confidence links	100	Conservative subset with composite score ≥ 0.70 and score margin ≥ 0.05 , <i>excluding</i> the 133 single-candidate links (margin undefined). A broader count of 116 results when single-candidate links with score ≥ 0.70 are also included.
Single-candidate links	133	Linked contracts with only one surviving candidate after filtering; margin is undefined for these.
Ambiguous multi-candidate links	265 of 572	46.3% of multi-candidate events have a score margin below 0.05.
Main unlinked failure mode	339 of 395	Most unlinked contracts have no temporal partner in the search window.
Secondary failure modes	32 text, 32 CPV	Some eligible contracts had a temporal partner but failed semantic or CPV checks.

The confidence diagnostics are useful for sensitivity analysis, not as legal ground truth. A strict subset can bound the event definition, but it does not prove that the remaining links are false.

8 Biases and Limitations

Selection bias remains possible because contracts with no temporal partner cannot be linked even if a real-world renewal occurred outside the observable window. Buyer fragmentation can create false negatives, while generic or repetitive contract descriptions can create false positives. Missing or generic CPV reduces auditability. Contracts from late 2023 face compressed observation windows, and 2024 contracts are structurally excluded from the eligible denominator because their expected renewal window falls after study end. Manual validation is still required to estimate the precision of the proxy renewal event.

Table 6: Principal data quality risks

Risk	Evidence in current data	Modeling impact
Buyer fragmentation	Post SIREN enrichment, 660 of 1,100 eligible rows carry a SIREN: key and 440 a NAME: key; 23 buyer merges added 8 renewal events (697 $\rightarrow$ 705). Residual name-based keys may still fragment some buyers.	Residual structural false negatives when the same buyer appears under multiple name spellings not resolved by the SIREN API.
Generic or missing CPV	Raw CPV missingness is 1.9%; generic codes are capped at low similarity.	Lower match precision and weaker audit trail for some links.
Sparse amount field	Only 21.4% of source APPEL_OFFRE notices have a usable amount.	Amount cannot serve as a reliable matching signal or segmentation variable.
Administrative duration values	69.5% of observed source durations equal 12, 24, 36, or 48 months.	Declared duration should not be interpreted as realized renewal time.
Observation-window censoring	395 eligible contracts are right-censored; late cohorts have less time to reveal renewals.	Apparent non-renewal may reflect study-end truncation rather than true absence of renewal.
Proxy event definition	BOAMP contains no explicit renewal label.	Event quality depends on the linking rule and should be stress-tested with confidence diagnostics.

9 Event Variable: Scoring, Robustness, and Validation Status

Exact definition

event = 1 is assigned when the linking algorithm finds a later BOAMP **APPEL_OFFRE** notice from the same buyer within a window of [6, 72] months around the source contract’s expected end date, passing a hard **text_similarity** ≥ 0.20 filter, and achieving the highest composite score among all candidates considered. **event** = 0 is right-censored: no such successor was detected under the rule before the 2024-12-31 study end. Right-censoring does *not* mean the contract was not renewed; it means no identifiable BOAMP successor was found under the current chaining rule. The variable is a **proxy chaining result**, not a legally certified renewal and not a confirmed absence of renewal.

Scoring formula and acceptance threshold

For source contract i and candidate j the composite score is:

$$S(i, j) = 0.40 s_{\text{text}} + 0.25 s_{\text{cpv}} + 0.20 s_{\text{temp}} + 0.15 s_{\text{buyer}}$$

When CPV is missing for either notice, the CPV weight is redistributed proportionally: $S = (0.40 s_{\text{text}} + 0.20 s_{\text{temp}} + 0.15 s_{\text{buyer}}) / 0.75$. The **only hard acceptance threshold** in the baseline model is **text_similarity** ≥ 0.20 ; no composite floor is applied. The observed minimum composite score for accepted links is approximately 0.37, arising from the weighting formula applied to pairs that pass the text filter. Two post-hoc confidence tiers are defined for sensitivity analysis:

- Medium/high confidence: composite ≥ 0.50 — 497 of 705 events (70.5%);
- Strict high-confidence: composite ≥ 0.70 *and* score margin ≥ 0.05 , multi-candidate links only — 100 events (14.2%).

Threshold sensitivity

Table 7 reports survival model outcomes under seven increasingly strict event definitions, sourced from `validation_robustness/outputs/threshold_sensitivity_summary.csv`.

Table 7: Threshold sensitivity of the survival model. KM median in months; “—” = median not reached within the study window.

Scenario	Events	Rate	KM med.	Cox C
Baseline (all accepted links)	705	64.1%	48.2	0.654
Score ≥ 0.50	497	45.2%	52.8	0.645
Score ≥ 0.60	275	25.0%	—	0.623
Score ≥ 0.70	148	13.5%	—	0.620
Score ≥ 0.80	79	7.2%	—	0.535
Score ≥ 0.90	25	2.3%	—	0.550
Strict HC ($S \geq 0.70$, $\Delta \geq 0.05$)	116	10.5%	—	0.647

The absolute event rate and KM survival level are sensitive to the event definition. The Cox concordance index, however, stays above 0.59 across all seven scenarios, and the direction of both significant hazard ratios ($HR < 1$ for declared duration, $HR > 1$ for start year) is preserved in every scenario. The qualitative conclusions of the survival model are therefore robust to the choice of linking threshold.

Uncertain cases

The median composite score of the 705 accepted links is 0.555 (IQR 0.489–0.653). Of the 572 multi-candidate links, 265 (46.3%) have a score margin below 0.05 — the algorithm selected a winner but an alternative was nearly equally plausible. Removing all uncertain links (composite < 0.50 or margin < 0.05) reduces the analysis to 767 contracts and 372 events (48.5%), shifts the KM median to 54.8 months, and lowers the Cox concordance from 0.6541 to 0.601. The drop in concordance confirms that the uncertain links carry positive discriminative signal and do not appear to introduce dominant systematic bias.

Manual validation sample

A stratified 150-row sample has been drawn to support future record-level review (file: `event_validation/outputs/manual_validation_sample.csv`).

Table 8: Manual validation sample design (150 rows). Review not yet executed.

Stratum	Criterion	<i>n</i>
A — High confidence	<code>event=1</code> , strict HC (<code>high_confidence_strict</code>)	20
B — Ambiguous margin	<code>event=1</code> , margin < 0.05 , multi-candidate	20
C — Single-candidate	<code>event=1</code> , <code>single_candidate_match=True</code>	15
D — No successor	<code>event=0</code>	50
E — Remaining linked	<code>event=1</code> , medium/low confidence, not in A–C	45
Total		150

Record-level review has not yet been completed. No empirical precision or recall estimate for

the linking step is therefore available. Until the review is done, the quality of the event variable rests solely on algorithmic confidence diagnostics and the threshold sensitivity analysis above.

Bias impact on survival analysis

Covariate balance between linked (event= 1) and censored (event= 0) contracts is good on all survival-model predictors: the standardised mean difference (SMD) is 0.128 for declared duration, -0.029 for start year, and -0.019 for the imputation flag — all well below the conventional 0.20 threshold for substantial imbalance. The largest observed imbalance is amount non-missing ($\Delta = -0.099$): censored contracts are more likely to carry a reported amount, possibly because high-value contracts are renegotiated through channels that do not produce a new BOAMP notice. This does not affect the survival model (amount is not a covariate).

Event rate by cohort year is broadly stable at 56–75% for contracts starting in 2015–2022. The 2023 cohort shows only 4 linked events out of 12 eligible (33.3%), but the sample is too small to interpret as a trend; the short observation window (less than 24 months to study end) is the more likely explanation. The dominant source of bias remains **structural right-censoring** of the 2022–2024 cohort, whose expected renewal windows fall after the 2024-12-31 study end — a study-design constraint, not a measurement error in the linking rule.

In summary, the 64.1% linking rate should be read as a **chaining result**: the proportion of eligible contracts for which the algorithm identified a plausible BOAMP successor under the current rule. It is not a measurement of the true renewal rate, because false positives and false negatives are both possible and no empirical precision estimate is yet available. The sensitivity analysis demonstrates, however, that the qualitative conclusions of the survival model are stable across all seven threshold scenarios tested. The variable is fit for purpose as a prioritisation signal, provided its proxy nature is communicated wherever the model output is used.

10 Reproducibility

This report is reproducible from tracked project outputs. The core files are:

- data/processed/boamp_full_clean.csv
- data/processed/boamp_phase2_survival.csv
- boamp_renewal_linking_quality/outputs/boamp_linking_stats.csv
- boamp_renewal_linking_quality/outputs/boamp_renewal_links.csv
- boamp_renewal_linking_quality/outputs/boamp_renewal_candidates.csv
- boamp_renewal_linking_quality/outputs/boamp_bias_report.csv
- boamp_renewal_linking_quality/outputs/boamp_link_confidence_diagnostics.csv

The relevant pipeline steps are implemented in `scripts/task_boamp_full_clean.py`, `scripts/task9_boamp_phase2_handoff.py`, and `scripts/task10_link_confidence_diagnostics.py`. No manual, untracked data manipulation is required to reproduce the counts reported here.

11 SIREN/SIRET Enrichment of Buyer Identifiers

11.1 Motivation: buyer fragmentation

The baseline pipeline builds a canonical buyer key using a SIRET-first rule: when a valid 14-digit SIRET is present in the BOAMP record, the key is `SIRET:{14-digit}`; otherwise it falls back to a normalized name key `NAME:{normalized_name}` after stripping accents, punctuation, and legal-form

suffixes. In the cleaned corpus, approximately 95% of rows carry a name-based key. Because the same public buyer can appear under slightly different spellings across procurements (*e.g.* “Ville de Nantes” vs. “Nantes Métropole”), name-based keys fragment what is in reality a single buyer into multiple groups. Fragmented groups reduce the pool of candidate renewals available to the linker and can produce false negatives in the event variable.

11.2 Data source and tool

Enrichment uses the *API Recherche d’Entreprises* (<https://recherche-entreprises.api.gouv.fr>), the official open-data search interface for the French SIRENE register. It is publicly accessible, requires no authentication, and covers all legal entities registered in France, including public bodies (*communes, départements, établissements publics, hospitals, intercommunalités*). The API returns up to ten candidates per query with a declared rate limit of 7 requests per second; the pipeline uses a conservative limit of 5 requests per second with local caching to avoid redundant calls on re-runs.

11.3 Why SIREN, not SIRET

SIRET (14 digits) identifies a specific establishment of a legal entity. SIREN (9 digits) identifies the legal entity itself. For buyer grouping, SIREN is more appropriate: a single public buyer (*e.g.* a *département*) can have multiple establishments with different SIRETs but a single SIREN. Using SIREN as the enriched key prefix (**SIREN:{9-digit}**) makes the source and grouping level unambiguous and distinguishes enriched keys from BOAMP-sourced keys (which retain the **SIRET:** prefix).

11.4 Matching method and confidence levels

For each unique buyer name, the pipeline:

1. Detects the likely entity type from the name (commune, SDIS, CHU, région, EPCI, etc.) to select the appropriate *nature juridique* API filter and query formulation.
2. Queries the API with the normalized buyer name, filtered by department code when available.
3. Scores every returned candidate using a token-sort string similarity (rapidfuzz) on normalized names, with a small bonus for token overlap and a department-match check.
4. Classifies the best candidate into one of four confidence levels:

Level	Score threshold	Interpretation
HIGH	≥ 85 , or ≥ 75 with single candidate or margin ≥ 10	Strong match; used automatically for buyer_key_enriched
MEDIUM	≥ 70 (not HIGH)	Plausible but ambiguous; retained for manual review only
LOW	≥ 55 (not MEDIUM)	Weak or generic match; not used automatically
NO_MATCH	< 55 or no candidates	No reliable candidate found

Table 9: Confidence classification for SIREN enrichment.

11.5 New buyer key

A new column **buyer_key_enriched** is defined with a simple rule: use **SIREN:{siren}** when the match is HIGH confidence and the SIREN is valid; otherwise keep the original **buyer_key**. MEDIUM, LOW, and NO_MATCH rows are therefore unchanged. The original **buyer_key** is preserved in **buyer_key_original**. Confidence levels MEDIUM and LOW are surfaced in **buyer_siren_enrichment/outputs/ambiguous_buyer_matches_for_review.csv** for optional manual validation but are never applied automatically.

11.6 Coverage and effect on renewal linking

To evaluate the effect of SIREN resolution, an **auxiliary Jaccard-based comparison** was run separately from the official Sentence-Transformer phase-2 pipeline, using `buyer_key_enriched` for grouping. This comparison uses the legacy Jaccard parameters (window ± 12 months, Jaccard threshold 0.20, weighting 0.7 text + 0.3 time) and produces two auxiliary files: `boamp_full_survival.csv` (name-based baseline, 1,933 rows) and `boamp_full_survival_enriched.csv` (SIREN-enriched, 1,933 rows) in `boamp_renewal_linking_siren_enriched/outputs/`. These are **legacy comparison artifacts**, not the current-stage survival dataset. Numerical comparison results are also saved in `buyer_siren_enrichment/outputs/baseline_vs_siren_enriched_linking_comparison.csv`. The observed effect was a modest increase in the Jaccard event rate, driven by merging contracts from the same legal entity that were previously split across multiple name keys.

11.7 Limitations

- **API results are not ground truth.** The SIRENE register records the entity as legally registered, not necessarily under the name used in procurement. Fuzzy matching can produce false positives, particularly for common generic names.
- **MEDIUM and LOW matches are not applied automatically.** They are kept in the review file for optional human validation.
- **HIGH-confidence mistakes remain possible.** A score ≥ 85 does not guarantee a correct match; it reflects strong textual similarity under a specific normalization scheme. Rare false positives at HIGH confidence would introduce a small number of incorrect group merges.
- **NO_MATCH buyers are unresolved.** Buyers with acronyms, internal codes, or abbreviated names (*e.g.* “CNIEG”, “SAELSI”, “DIR Ouest”) often fail to match and remain name-keyed. Their renewal groups are unchanged relative to the baseline.
- **The SIREN enrichment annotation does not recompute the event variable.** The official phase-2 dataset (`boamp_phase2_survival.csv`, 705 events) incorporates SIREN-based buyer key merges into its `buyer_key` column (660 SIREN: keys, 440 NAME: keys). The file `boamp_phase2_survival_siren_enriched.csv` extends it with four annotation columns (`buyer_key_enriched`, `enriched_siren`, `enrichment_confidence`, `enrichment_score`) added by left-join; these columns record enrichment metadata but do not change the event or survival-time columns.

Scientific framing. SIREN enrichment is a buyer-resolution improvement, not a new ground truth. It is expected to reduce false negatives arising from buyer fragmentation and to improve the precision of renewal-group boundaries, but it cannot compensate for the fundamental limitation that **event** remains a proxy.

12 Conclusion

The BOAMP dataset satisfies the internship guide’s minimum data-quality requirement for Phase 1 because it provides a documented corpus, explicit cleaning rules, missingness information, a transparent linking rate, and a survival-ready population of more than 800 eligible contracts with either observed or censored duration. The main limitation is conceptual rather than technical: **event** remains a proxy renewal event, not legal ground truth, so later manual validation is still needed to estimate precision.